TopoLight 0.1

Technical datasheet

Self-hosted network monitoring with a live LLDP/CDP topology map · one static binary · offline licence

Edition covered: 0.1.x (Free, Pro, Team share one binary). Numbers marked "measured" come from the 1,500-device simulated estate used for release testing; nothing else is estimated.

Self-hosted network monitoring with a live LLDP/CDP topology map. One static binary, embedded storage, offline licence. This sheet lists exactly what TopoLight collects, what it shows, what it needs, and what it does not do yet — so you can compare it with other tools on facts.

1. Data sources and protocols

Source	What TopoLight collects	Notes and limits
ICMP echo	Round-trip time, packet loss, jitter per device, every poll cycle (default 60 s)	Raw sockets on Linux (`cap_net_raw` set by the installer, or the unprivileged ping sysctl). macOS/Windows builds run SNMP-only reachability.
SNMP v2c	Read-only community polling and discovery	Community stored on disk (mode 0600), never returned by the API.
SNMP v3	authPriv / authNoPriv / noAuthNoPriv; auth SHA-1, SHA-256, MD5; priv AES-128, DES	Own BER codec and USM implementation, tested against net-snmp.
SNMPv2-MIB / HOST-RESOURCES	sysDescr, sysObjectID, sysUpTime (reboot detection), sysName, sysLocation; hrProcessorLoad and hrStorage as CPU/memory fallback	Reboot = uptime went backwards *and* new uptime < 24 h (no false alarms on counter wrap).
IF-MIB / ifXTable	Per interface: name, alias, type, speed, MAC, admin/oper status; 64-bit octet counters → bit/s and utilisation; unicast packet counters → packets/s; ifInErrors/ifOutErrors → errors/s; ifInDiscards/ifOutDiscards → drops/s	32-bit counters used when the agent has no ifXTable, with wrap handling. Interface kinds (physical, LAG, VLAN, tunnel, loopback) classified from ifType.
ENTITY-MIB	Chassis serial number and model	Used for inventory and for LLDP chassis matching.
Vendor MIBs (built-in profiles)	CPU, memory, temperature, sessions: Cisco IOS/IOS-XE, NX-OS, ASA/Firepower; Fortinet FortiGate; Palo Alto; Juniper; Aruba AOS-S and AOS-CX; MikroTik; Huawei; Ubiquiti; net-snmp hosts	Profiles are JSON; drop your own in `<data>/profiles/` to add a vendor without touching the poller.
LLDP-MIB	Local port table, remote chassis ID/port ID/system name/description	Every observation is kept with a confidence score; a link seen from both ends scores 1.0, one end 0.8, an unknown neighbour 0.6.
Cisco CDP (CISCO-CDP-MIB)	Neighbour device ID, port, platform, address	Enabled by the Cisco profiles; feeds the same link builder as LLDP.
Syslog UDP + TCP (port 514)	RFC 3164 and RFC 5424 framing; Cisco `%FACILITY-SEV-MNEMONIC`; FortiGate `key=value`	Lines are matched to devices by source address; unknown senders are counted so you can find devices that talk but were never discovered. TLS syslog is not in 0.1.
SNMP traps and informs (v2c, port 162)	linkDown/linkUp, coldStart/warmStart, authenticationFailure, BGP and ENTITY notifications, plus a vendor-agnostic fallback that keeps raw varbinds	v3 traps are not in 0.1.
Outbound	SMTP with STARTTLS; Telegram Bot API; webhook with X-	Grouping window, quiet hours, minimum

Source	What TopoLight collects	Notes and limits
	`TopoLight-Signature: sha256=<HMAC>`	severity, "critical always" punch-through.

2. What it does with the data

Discovery. Sweeps subnets and ranges (largest /20 per line) with ICMP then SNMP using every saved credential; follows LLDP/CDP neighbours automatically; manual add for anything else. Devices beyond the licence cap are listed as *not monitored*, never silently dropped.

Topology. Links are synthesised from both ends of every LLDP/CDP observation. Roles (core, distribution, access, router, firewall) are inferred from the graph — a redundant core pair counts as two — and can be locked by hand. The map is a 3D stacked-disc view drawn on a plain canvas (no WebGL, no JavaScript framework) with a 2D mode, orbit, zoom, status and utilisation overlays, and live updates over a server-sent event stream. Links that disappear are kept greyed for 7 days, then dropped; manual links are never dropped.

Health at a glance. Hovering a node shows a health card: status and cause, CPU, memory, temperature, RTT and loss, interface counts (up / down / shut, uplinks down), aggregate traffic in and out as bit/s and packets/s, drops and errors per second, the three busiest ports, and the ports that are admin-up but operationally down. The same card is in the click-through panel and the device page.

State engine. Per-device and per-interface state machines with confirmation cycles (default: down after 3 failed cycles, up after 2 good ones), separate enter/exit thresholds so a value hovering at the line does not flap, flap detection (more than 5 changes in 10 minutes), and **topology-aware suppression**: when a device goes down, everything whose only path runs through it becomes *unreachable* and its alert folds under the root cause. More than 80 % of a site down collapses into one site alert. Maintenance windows silence devices without deleting anything.

Alert rules (all editable). device down, SNMP unreachable, CPU, memory, temperature, ICMP loss and latency, interface utilisation and error rate, link down (trap/syslog first, confirmed by a poll), reboot, configuration change, BGP/OSPF neighbour, HA state change, environment fault, authentication failures, log flood, VPN tunnel change. Resolved alerts that re-trigger within 30 minutes re-open with their history instead of duplicating.

Console. Overview, topology, alerts with keyboard navigation (`j`/`k`/`a`/`r`), device pages with 24-hour graphs, log explorer with histogram, admin for sites, credentials, notifications, rules, maintenance, users and licence. Dark and light themes, WCAG AA (0 axe violations on 18 page/theme combinations at release), command palette on `/`.

Security. Login required from the first request; PBKDF2-SHA256 (600k) passwords; HttpOnly + SameSite=Strict sessions; same-origin enforcement on state-changing requests; CSP forbidding inline and remote scripts; credentials never leave the data directory. TopoLight is read-only towards devices — it never pushes configuration.

3. Editions

	Free	Pro	Team
Monitored devices	25	500	1,500
Sites	1	3	unlimited
Metric and log retention	7 days	6 months	12 months
Users	1 admin	3 admins	unlimited; admin / operator / viewer roles
Everything in §1–2 (ICMP, SNMP v2c/v3, LLDP/CDP topology 2D + 3D, syslog + traps, state engine, e-mail)	✓	✓	✓
Telegram and signed webhook	—	✓	✓
Maintenance windows, JSON export API	—	✓	✓

	Free	Pro	Team
Price	$0 (GitHub)	$49 / month	$149 / month

One binary for all three; an offline Ed25519 licence key unlocks the caps. Over the cap, the newest devices are marked *not monitored* and the API answers `402` with a readable message — nothing breaks, nothing is deleted. A missing or expired key runs as Free.

4. Server requirements

TopoLight is **one process on one host**. The binary contains discovery, the ICMP pinger, the SNMP poller, the syslog and trap receivers, the topology builder, the state and alert engine, the notifier, the embedded time-series and log stores, and the web console. Nothing is installed on the monitored devices; no database server, message bus, cache or agent is installed next to it.

Measured on a 1,500-device simulated estate polled every 60 s: about 0.25 CPU cores on average and ~110 MB RSS. Disk for history is the real cost: **≈4.5 KB per series per day at 60-second resolution** and **≈1.8 KB per series per day at 5-minute resolution** (worst case, noisy values). Raw 60-second samples are kept 7 days, then 5-minute avg/min/max to the retention limit; non-uplink ports are stored at 5-minute resolution from the start.

Estate	Host	History on disk (worst case)
100 devices, 24-port switches, 6 months	1 vCPU · 1 GB RAM	~2 GB
500 devices, 48-port switches (Pro, 6 months)	2 vCPU · 2 GB RAM	~17 GB
1,500 devices, 48-port switches (Team, 12 months)	2 vCPU · 4 GB RAM	~100 GB

Idle access ports compress far better than the worst case; real estates typically land at a third of those figures. Admin → System shows the live number.

Network. Outbound UDP/161 to devices; inbound UDP/514 + TCP/514 (syslog) and UDP/162 (traps) from devices; TCP/8432 for the console (put TLS or a reverse proxy in front, or use `-tls-cert`/`-tls-key`). Linux amd64/arm64 for production; macOS and Windows builds for trying it on a laptop.

Deployment modes

Mode	Status in 0.1	What it means
Standalone	shipped	One host monitors the whole estate over routed IP. The only supported mode today.
Warm standby	works, manual	Hypervisor HA or a scheduled copy of `/var/lib/topolight` to a second host; the data directory is self-contained, so a restore with the same key brings history back. Point syslog/trap targets at a movable VIP or DNS name.
Active/active HA	not yet (0.2)	Two instances today would double SNMP load and duplicate alerts; no shared state or leader election in 0.1.
Remote collectors / cluster	not yet (0.2)	A lightweight collector per site for many-site, NAT'd or >1,500-device estates.

5. How it compares

Stated as differences, not verdicts — each of these tools is good at things TopoLight does not attempt.

	TopoLight 0.1	**LibreNMS**	**Zabbix**	**PRTG**
Install footprint	1 static binary, embedded stores, one directory to back up	PHP + MySQL/MariaDB + RRD + poller workers	Server + database (PostgreSQL/MySQL) + frontend + agents/proxies	Windows core server + probes
Topology	Drawn from LLDP/CDP with confidence scores; roles inferred; 3D/2D live map	Discovery-based maps and plugins	Manual/network maps, discovery rules	Auto-discovery, sensor tree, maps
Root-cause handling	Graph-based suppression built into the state engine (unreachable ≠ down)	Dependency-based parent/child	Trigger dependencies, configured per trigger	Sensor dependencies
Traffic data	SNMP counters (bps, pps, drops, errors)	SNMP + NetFlow/sFlow via plugins	SNMP + collectors	SNMP, flow, packet sniffer sensors
Pricing model	Per instance: 25 devices free, $49 (500), $149 (1,500); offline key	Free (GPL)	Free (AGPL) + paid support	Per sensor
Breadth	Network devices only, on purpose	Very broad	Very broad (servers, apps, cloud)	Very broad

6. Not in 0.1 — public roadmap

NetFlow/IPFIX/sFlow collection · gNMI/OpenConfig streaming · SNMPv3 traps · syslog over TLS · MAC/ARP endpoint placement · TCP/HTTP/DNS/TLS synthetic probes · BGP/BMP peering · configuration backup · active/active HA and remote collectors · API tokens · Docker Hub image. Against a full enterprise NMS/observability catalogue (availability, health, flow, packet, L2 topology, routing, streaming telemetry, logging, identity, network services, wireless/SD-WAN, OAM, cloud) TopoLight 0.1 covers availability, topology and device/interface performance well, event logging partially, and none of the rest — that is the scope of a tool built for 25–1,500 network devices, not a platform claim.

7. Links

GitHub (free edition, source, issues): github.com/nizartuanku/topolight · Whop (Pro/Team, 14-day trial): whop.com/nizar-tuanku/topolight · Docs: `docs/INSTALL.md`, `docs/USER-GUIDE.md`, `CHANGELOG.md`. Part of the Hexward line of self-hosted tools.